

TEACHING STATEMENT - LUCAS GAGNON

I teach by giving students broad context, specific know-how, and engaging problems that build mastery. I create a collaborative course culture to catalyze deep and meaningful learning. I believe that all students can learn, and effective teaching helps students become better learners.

This statement will expand on my teaching philosophy and outline my goals, drawing from my experience at the University of Southern California (USC), York University, and the University of Colorado Boulder (CU). This includes:

- teaching applied combinatorics at USC, an upper-division course for math majors,
- teaching theoretical linear algebra and introduction to proofs at York, which are both lower-division courses for math and related majors, and
- teaching precalculus at USC, differential calculus at CU and York, integral calculus at CU, and multivariable calculus at CU, which had students from many STEM fields.

I have also mentored or co-mentored eleven undergraduate researchers in five projects:

- an MSRP at USC on combinatorial polynomials, summer projects at York on Hopf algebras and machine learning, and an REU and Honors project at CU, both on group representation theory.

Through these experiences I have developed and implemented a blend of active and engaging lessons, rigorous assessment, and positive social environments that build understanding and encourage student growth.

Active lessons: I believe that students build mathematical understanding by cyclically assimilating new ideas, directly applying them, and reflecting on their experience. This leads me to mix lecture and active learning strategies in my classroom. I will illustrate this approach by with examples from a recent lesson about the “partition type formula” for Stirling numbers $S(n, k)$ of the second kind that I taught in Applied Combinatorics at USC.

I begin by activating students’ curiosity and contextualizing learning goals, often with an interactive lecture or group discussion about “what do we already know, and what are we unsure about?” This draws students in while informally assessing preparedness for the new topic. In my lesson at USC, I recalled a previous formula for $S(n, k)$, the “multiset formula,” and introduced the partition type formula as a computational black box, paired with the question: which formula is better?

I then transition to individual reflection or small-group discussion focused around a task, which can range from a problem written on the board to a tactile of computer interactive. During this time I conduct check-ins, asking questions like “how do you know this?” and “why?” In my lesson at USC, I asked groups to compute $S(6, 4)$ using both formulas in order to decide which formula was better, and through check-ins I was able to help several

students implement the formulas, while noticing that other students had advanced to discuss an interpretive proof of the new formula.

After a short interval, I bring the class into a summarizing group discussion, where I try to engage a wide range of students. This reinforces the students' role as knowledge creators and builds confidence and mathematical communication skills. At USC the discussion was framed around the challenges and benefits of using each formula, and the class identified a number of abstract qualities (ease of computation, interpretability, and simplicity) by which any formula could be judged. One group shared their insight about proving the partition type formula, and another performed critical thinking by asking how the number of terms in each formula compared outside of the $S(6, 4)$ example.

With the students engaged and up to speed, I transition into a detailed lecture, where I model mathematical reasoning and communication by naming key concepts, framing logical arguments, and writing grammatically complete sentences. In my lesson at USC, I expanded on my students' insight with a rigorous proof of the partition type formula with the division counting principle. I emphasized the meaning of the formula, identified the summands as discrete objects, and verified the setup and hypotheses of the division principle.

I conclude each lesson by drawing connects to our objectives and framing questions and building connections outside the classroom. At USC, I emphasized the importance of judgement in math, previewed our upcoming lesson on counting partitions as an answer to the lingering number-of-terms question, and situated the lesson in the active and ongoing scholarly debate about the meaning of "combinatorial formula."

Assessing student learning: In order to encourage wider proficiency and avoid bias, I conduct summative assessment beyond in-class exams to measure student learning. For applied combinatorics, I have paired low-weight weekly homework problem sets with two 10-minute presentations on these problems; in previous classes I have used low-stakes quizzes and revision-based mastery portfolios to encourage fluency and mastery. Changing technology—from integral calculators to ChatGPT—require regular reflection and evolution of these methods. While technological development changes the viability of particular methods, but I find they produce specific improvements over the course of the semester, such as increased confidence in mathematical discussions, improved fluency, or deeper metacognition.

When writing an exam or assignment, I identify specific learning objectives and write questions and rubrics to measure student's understanding of these objectives. I began this practice as Assistant Calculus III Coordinator at CU in order to reduce bias and improve assessment accuracy. This practice also gives students with clearer feedback, making it possible to tell them precisely which learning goals they achieved and where they may have fallen short. I also try to make exams enjoyable for students, interting levity and meaning into exams through carefully-considered examples and instructions.

Classroom community: I cultivate supportive peer connections and collaborative learning in my classroom; my goal is to create a welcoming and inclusive culture. Research shows

that peer connections improve student belonging and builds long-term peer networks, and I find that it also leads to more collaboration, buy-in, and accountability in my classroom. Implementing concrete strategies accomplish this, such as introducing roles in group discussion, was a key part of my role in the Student Academic Success Center (SASC) at CU and I continue to use this practice to teach more equitably and effectively.

My approach to building community is cumulative. In our first meeting, I work with students to describe norms and behaviors that make it possible to ask questions and take risks; at the same time I form well-mixed collaborative peer groups in class. As the semester progresses, I encourage students to seek out new working partners and model collaboration and risk taking, as outlined in our first-day norms. As I notice students proactively discussing and collaborating, I ask all members of the class to engage in higher-stakes exercises such as presentations, participating, and structured peer review.

Mentorship: Beyond classroom teaching, I have experience supervising undergraduate and graduate researchers and teaching assistants. In this capacity, I work with each student to identify our common objectives alongside the student's individual goals. I then scaffold research projects and teaching tasks around this shared agreement. By combining supporting feedback and skill-appropriate challenges, this helps students develop as emerging mathematical contributors.

Another component of mentoring and supervisory work is producing concrete evidence of student's expertise such as publications and project reports. This has been an area of growth over time: I recently published a paper with undergraduate researchers at York, and am preparing a manuscript with another group based on a summer MSRP at USC. In mathematics pedagogy, I helped two undergraduate teaching assistants at CU develop their own active learning worksheets, which were contributed to a curriculum repository.

Growth: I am still early on my pedagogical journey, and I continue to grow by teaching new courses, developing new lessons and assessments, watching colleagues teach, and soliciting mid-semester feedback from my students. I have also completed intensives at York and CU focused on college teaching, and have produced a portfolio of lesson plans and teaching notes.

I also grown though reflection on my current teaching: many the practices described in this statement were shaped by prior challenges. For example I found early in my career that unstructured group discussion brought out the same confident voices and responded by introducing individual preparation time and whole-class synthesis, which broadened participation. I will continue to adapt and improve as I introduce new strategies, evaluate their success, and identify their drawbacks.

Conclusion: Although I continue to learn, my experience makes me comfortable using active learning in many kinds of classroom. I am committed to equitable teaching practices and cooperative learning. I am excited to explore new challenges and approach excellence in all aspects of teaching.